GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for Bolts and Nuts**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-012
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Originator	Emmanuel Uwaoma
Security Classification	Restricted
Issue Date	**27-Jul-22**

Specification for Bolts and Nuts

R01	27.07.22	Issued for Review	E. Uwaoma	E. Nwachukwu	A. Duru	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev No.	Date	Description of Revision
R01	27/07/22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-012

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviations	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
AWS	American Welding Society
CS	Carbon Steel
DEP	SPDC- Design and Engineering Practices
DP	Design Pressure
DWT	Drop Weight tear
FEED	Front End Engineering
HAZ	Heat affected zone
FCAW	Flux Cored Arc Welding
HV 10	Vickers Hardness (10 kg load)
GMAW	Gas Metal Arc Welding
GTAW	Gas Tungsten Arc Welding
ISO	International Organisation for Standardization
IF	Incomplete Fusion
IFD	Incomplete Fusion Due to Cold Lap
IIW	International Institute of Welding
IP	Inadequate Penetration Without High-Low
IPD	Inadequate Penetration Due to High-Low
IQI	Image Quality Indicator
MMscf/d	Million standard cubic feet per day
NDT	Non Destructive Testing
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SPDC	Shell Petroleum Development COMPANY Nig. Limited
SMAW	Shielded Metal Arc Welding
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

Site and Environmental Conditions

Refer to Basis of Design Document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

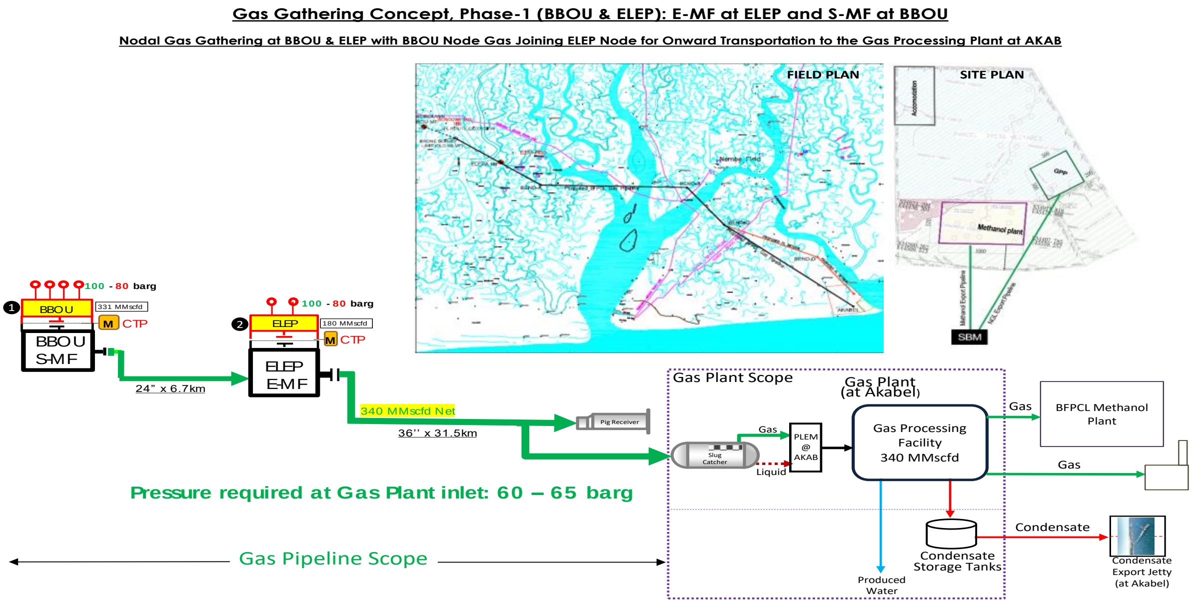

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification describes the technical requirements for design, production, coating, testing & inspection, storage and delivery of Bolts and Nuts that shall be used for piping installations in the BFPCL Gas Gathering Pipelines (Phase1).

This specification may not specify all details for manufacturing as would be required for an adequate supply. It is the manufacturer's responsibility to ensure that all supplied bolts and nuts have been appropriately manufactured and tested in accordance with relevant codes, standards

2. Regulations, Codes and Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- COMPANY's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the COMPANY for resolution.

2.1. Project Technical Documents

REF	DOCUMENT NUMBER	TITLE
[1]	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
[2]	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Classes
[3]	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

2.2. Shell Specifications

The following DEPs are applicable:

REF	DOCUMENT NUMBER	TITLE
[4]	DEP 31.38.01.25-Gen.	Piping – Process Design Requirements
[5]	DEP 31.38.01.24-Gen.	Piping – Engineering And Layout Requirements
[6]	DEP 30.48.00.31-Gen	Protective Coatings For Onshore And Offshore Facilities
[7]	MESC SPE 81 Group	Bolting
[8]	MESC SPE 76/101	Flanges (Amendments/Supplements To ASME B16.47)
[9]	MESC SPE 76/100	Flanges (Amendments/Supplements To ASME B16.5)

2.3. International Codes and Standards

REF	DOCUMENT NUMBER	TITLE
[10]	ASME B1.1	Unified Inch Screw Threads (UN and UNR Thread Form)
[11]	ASME B16.5	Pipe Flanges and Flanged Fittings
[12]	ASME B31.3	Process Piping
[13]	ASME B 1.2	Gages and Gaging for Unified inch Screw Threads

[14]	ASME B 18.2.1	Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
[15]	ASME B 18.2.2	Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
[16]	ASTM A 193	Standard specification for Alloy-Steel and Stainless Steel Bolting Materials for High-temperature or high pressure service
[17]	ASTM A 194	Carbon and Alloy-Steel Nuts for Bolts for High-pressure or High-temperature Service
[18]	ASTM A 370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products

3. Bolting Requirements

3.1. General

Stud-Bolt design shall be as per ASME B18.2.1

Stud-bolts shall be designed to have full continuous rolled threads along its entire length with two heavy hex nuts per bolt. Square head machine bolts shall not be supplied.

All nuts shall be of the semi-finished hexagonal type ("Heavy" series) and shall be in accordance with the standard ASME B18.2.2.

3.2. Threading

Threading on studs and nuts shall be in accordance with ANSI B1.1 and the following classification:

- sizes 25 mm (1 in) and smaller to be Coarse Thread Series (UNC);
- sizes larger than 25 mm (1 in) to be 8-Thread Series (8 UN);

The machining and surface-condition tolerances shall be in conformity with the following classes:

- Class 2A of standard ASME B1.1 for stud bolts
- Class 2B of standard ASME B18.2.2 for nuts.

3.3. Dimensions

3.3.1. Stud bolts and screws

The diameters and lengths of stud bolts shall comply with the standard ASME B16.5

Diameters shall be in inches, and lengths shall be in millimeters.

The lengths given on the requisitions and MTO are service lengths as per ASME B16.5 and do not include the length of the end bevels.

Studs of 50 mm (2 in) diameter and larger shall be the larger of one nut thickness or 40 mm (1 ½ in) longer than normal to permit use of a bolt tensioner as per DEP 31.38.01.24-Gen. Ref [4]

3.3.2. Nuts

Generally, nuts shall have a height equal to the bolt diameter.

3.4. Materials Requirements

Refer to Ref [2] piping material classes for selection of material standards for bolts and nuts.

Bolts materials for moderate and high temperature non-sour services shall be in accordance with ASTM A193 and ASTM A194 for Nuts for Carbon Steel and ASTM A453 Grade 660 Class B for Duplex Steel.

In accordance with ASME B31.3, for non-insulated components with fluid temperatures ≥ 65 °C (150 °F) the bolting temperature shall not be less than 80 % of the fluid temperature.

In order to avoid tightness failure due to temperature effect, a temperature limitation has been placed on the bolting material combinations in relation to the applied piping material in the table below:

Piping Material	Bolt Material spec	Nut material Spec	Allowable Temp.	Diameter Range
Carbon steel	A193-B7	A194-2H	-40 °C to 410 °C	2½ ≤ d ≤ 4
LT Carbon steel	A320-L7	A194-7	-73 °C to 343 °C	½ ≤ d ≤ 4
Duplex Steel	ASTM A453 Grade 660 Class B	ASTM A453 Grade 660 Class B	-40 °C to 410 °C	2½ ≤ d ≤ 4

When grade 7 of A194 nuts is specified for low temperature service, supplementary requirement S3 of ASTM A194 Ref [16] shall apply.

3.5. Fabrication

Stud bolts and nuts threads shall be machine cut or rolled. However, nuts may be forged.

Repair welding of bolting is prohibited.

After the operations of forging and machining, the bolting shall undergo the hardness treatments required by the reference codes and standards. Mechanical property requirements for studs shall be the same as those for bolts

3.6. Coating Bolts

External surfaces of bolting shall be as per described in the piping classification or material requisition.

3.7. Marking

Alloy steel and stainless-steel bolting shall be marked in accordance with ASTM A193 while carbon and alloy steel nuts shall conform to ASTM A194. And

Stud bolts and Nuts shall be marked with the following data:

- Manufacturer Stamp
- Steel type and grade

3.8. Quality control

Bolts and nuts shall be visually inspected without magnification and confirmed to be free from bum, seams, laps, loose scale, irregular surfaces, and any defects affecting their serviceability.

The inspection and tests necessary to confirm that the products meet the requirements of the standards, specifications and requisitions shall be carried out in the Manufacturer's facility. This specially-qualified personnel shall be independent of the production department of the plant.

Products shall be guaranteed by an inspection certificate (conformity, material, test, etc.), which shall be submitted by manufacturer.